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Textural controls on low-frequency electrical spectra of porous media

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ABSTRACT

The results from several laboratory studies of the relationships between electrical polarization and physical properties of porous media have prompted interest in the potential use of low-frequency electrical spectra to qualitatively or quantitatively map variation in hydrogeologic properties in the field. Compiling several published and unpublished data sets, supported by new measurements, we have examined the low-frequency electrical spectra of a range of natural and artificial porous media to assess the generality of proposed relationships between electrical and physical properties. Our work confirms a significant positive correlation between the magnitude of electrical polarization (quantified as imaginary conductivity at a specific frequency) and the surface-area/pore-volume ratio S_{por}. Analyzing the parameters of a generalized Cole-Cole resistivity relaxation model fitted to many electrical spectra, we observe two apparent controls on the electrical relaxation. For samples with abundant relatively large pore throats, we observe a distinct increase in the time constant of the model with modal pore-throat size, in accordance with classical electrical relaxation models. However, for media with pore structures dominated by small pore throats, the diffusion-length scales do not appear to be controlled by modal pore-throat size. We conclude that for such media, the microstructure of the network of small pores leads to some connectivity of diffusion paths; thus, these samples exhibit relatively large time constants. There is potential value in addition to limitations when using electrical spectra to estimate physical properties of porous media, and we see the need for more appropriate generalized theories of electrical polarization in hydrogeologic media.

INTRODUCTION

One of the most important and challenging goals in hydrogeophysics is the ability to predict reliably the hydraulic properties of subsurface formations. Hydraulic conductivity often is considered a critical parameter for hydrogeologic conceptual model development and, consequently, hydrogeologic decision-making. For some cases, an assessment of the spatial variability of this property is also important, particularly in solute-transport studies. Other properties, such as porosity and water-retention characteristics (for unsaturated-zone studies), also are often required. As we attempt to understand the fate of dissolved chemicals migrating in the subsurface environment, the properties that influence the mobility of solutes and their interaction with geochemically active mineral surfaces also are important for site characterization and model development.

All of these properties are difficult to measure in situ with conventional approaches. Measurements through retrieved core samples are limited because of the small measurement support volume and inevitable disturbance of the porous media in the sampling process. These constraints have led to several studies of geophysical techniques for hydrogeologic-parameter estimation. Geophysical well logging remains a useful support for hydrostratigraphic logging in boreholes, but if we could reliably quantify key hydraulic properties from geophysical logs, then significant advances in hydrogeologic characterization would result. Even greater advances would be achieved if we could map or image hydraulic properties through geophysical surveys using ground-surface or in-well sensors (e.g., Kemna et al., 2004).

Hydraulic and electrical storage and transport are functions of the grain and pore microgeometries, pore connectivity, and solution chemistry in earth materials. Consequently, electrical methods are used widely in hydrogeophysical investigations to obtain high-resolution information about subsurface conditions. Electrical methods include electrical resistivity, induced polarization (IP), complex resistivity, EM induction, and ground-penetrating-radar (GPR) surveys. These methods, which operate at frequencies ranging from di-

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rect current (DC) to >1 GHz, can be used individually or in combination to obtain information about the subsurface structure and composition.

This apparent link between hydraulic and electrical properties has encouraged many studies on predicting key hydraulic properties, such as hydraulic conductivity, from electrical measurements. Relationships between DC electrical conductivity σ and hydraulic conductivity K have been explored in a range of porous media. In clay-free soils, a positive correlation between σ and K may be apparent because the electrical conductivity will be strongly (and positively) influenced by porosity Φ. In clay-rich soils, however, such a relationship is unlikely to hold because the grain-surface electrical conductivity caused by the clay will increase the bulk electrical conductivity.

It would appear, therefore, that universal models explaining the relationship between σ and K are unlikely to exist (Purvance and Andricevic, 2000). However, the ambiguity in such relationships may be reduced significantly by incorporating charge storage within a petrophysical model because this will be influenced by characteristics of the pore/grain interface. Charge storage can be measured at one current-injection frequency or at a range of frequencies (spectral induced polarization). Here, we use the term *complex conductivity* (or its inverse, *complex resistivity*) to refer to a single frequency measurement of electrical conductivity and charge storage (i.e., the in-phase and quadrature components).

Given that charge storage and conductivity are affected by properties of the pore/grain interface and the pore space, useful relationships between electrical spectra and a range of hydraulic and geochemical properties may exist. In fact, relationships between parameters determined from the characteristic shape of electrical spectra measured on nonmineralized, ionically conducting rock samples and several intrinsic properties such as specific surface, cation exchange capacity (CEC), clay content, and grain or pore size have been suggested over the past three decades. Klein and Sill (1982) report on a study using artificial mixtures of clay and glass beads. They observe that the electrical spectra change with average grain size of the artificial mixtures. Lesmes and Morgan (2001) also offer support for relationships between grain-size distributions and the electrical spectra of sandstones. Börner et al. (1996), working with a range of sandstones, develop a model between properties of the electrical spectra and hydraulic conductivity using their empirical relationship between complex resistivity and surface area of the porous media. Studying unconsolidated sieved sand mixes, Titov et al. (2002) show experimental evidence of a link between grain size and the square root of the relaxation time of polarization. Recent electrical spectra measurements on a variety of sandstones from the U.K. and Germany (Scott and Barker, 2003; Binley et al., 2005) show that the length scale controlling the electrical relaxation in brine-saturated sandstones is likely to be associated with pore throats in the sandstone.

Although these examples in the literature reveal the potential of using electrical spectra for estimating key hydrogeologic properties, the universality and nonuniqueness of these relationships have not been explored. Here, we attempt to address this gap in knowledge by evaluating controls on the shape of electrical spectra for a range of consolidated porous media. In particular, we assess the generalization of previously proposed relationships between (1) the relaxation of electrical spectra and characteristic pore size and (2) the electrical polarization and surface area. By considering other potential influences on electrical spectra, we assess the potential universal applicability of hydrogeophysical relationships based on electrical spectra. Our study uniquely brings together data sets from several recent published and unpublished studies and is supported by new observations on a number of samples.

We review the empirically derived Cole-Cole class of models and explain how we use them to fit the complex conductivity measurements. In particular, we use the time constant from the Cole-Cole model fits as a measure of the characteristic diffusion time and relate this parameter to a characteristic pore or grain size for the suite of samples that we have assembled. In this way, we hope to evaluate the universality of the relationship between the spectral IP response and the intrinsic microgeometric properties of porous materials. We concentrate on the four-electrode, 10^{-3}–10^{3}-Hz frequency range because within this range distinctive, peaks in electrical spectra often are observed and because extension of these measurements at the field scale is, at least theoretically, possible (albeit challenging).

ELECTRICAL PROPERTIES

The conductive and capacitive properties of a material can be represented by complex conductivity σ^*, complex resistivity ρ^*, or complex permittivity ε^*, where

$$\sigma^* = \frac{1}{\rho^*} = i\omega\varepsilon^*. \quad (1)$$

Here, ω is the angular frequency ($\omega = 2\pi f$) and $i = \sqrt{-1}$. The complex electrical parameters can be expressed in polar or Cartesian form. For example, the complex conductivity can be expressed in terms of a magnitude $|\sigma|$ and phase ϕ or by real σ' and imaginary σ'' components:

$$\sigma^* = |\sigma|e^{i\phi} = \sigma' + i\sigma''. \quad (2)$$

Whereas σ' principally depends on the saturating-fluid conductivity and porosity of a porous medium, σ'' is influenced by the characteristics of the electrical double layer at the matrix/fluid interface. For ionically conducting porous media, the real and imaginary components of the complex conductivity vary with frequency. For illustration, the complex-conductivity response of Berea sandstone saturated with 0.01 M sodium chloride (NaCl) and measured over the frequency range of 10^{-3}–10^{6} Hz is plotted in Figure 1.

Diffusion polarization mechanisms

Although many theoretical models have been proposed to explain the complex-conductivity response of ionically conductive porous media, there is no universally accepted model that can explain all of the observed behaviors.

The complex-conductivity response of rocks and soils is controlled by physical and chemical polarization mechanisms that are intrinsically coupled and that operate over a broad range of frequencies. Experimental and theoretical evidence indicates, however, that the dispersion in the low-frequency complex-conductivity response is controlled primarily by diffusive polarization mechanisms (e.g., Chelidze and Gueguen, 1999; Chelidze et al., 1999). Purely physical polarization mechanisms (e.g., Maxwell-Wagner polarization) are unable to reproduce the large imaginary conductivity responses observed at low frequencies (e.g., Lesmes and Morgan, 2001), which result in a relatively constant phase response at low frequencies (see Figure 1).

Diffusion polarization results when an electric field is applied to a porous material and ions migrate through the bulk pore solution

along grain-solution interfaces and across blockages at pore throats (e.g., clay blockages). While migrating through and across these different regions, the ions travel with varying mobilities and accumulate at physical/chemical discontinuities, thus storing charge (i.e., a low-frequency capacitance). When the applied electric field is removed, the accumulated ions diffuse back to their neutral equilibrium distribution. The time that it takes the ions to reach equilibrium depends upon the distance over which the ions are polarized and their effective mobility. For the polarization of spherical particles (e.g., grains) of radius r, the relaxation time τ_s is given by (e.g., Schwarz, 1962; Schurr, 1964; Fixman, 1980)

$$\tau_s = \frac{R^2}{2\mu_s kT} = \frac{R^2}{2D_s}, \tag{3}$$

where μ_s is effective ionic mobility, k is Boltzmann's constant, T is absolute temperature, and $D_s = \mu_s kT$ is the effective diffusion coefficient.

If we assume, for simplicity, that the diffusion polarization process is strictly controlled by the pore-size distribution, then smaller pores will lead to faster relaxation than larger pores and, in terms of frequency, smaller pores will polarize at higher frequencies and larger pores will polarize at increasingly lower frequencies. If the complex-conductivity response is measured over a sufficiently wide range of frequencies, it should be possible to relate the shape of the complex-conductivity spectra to the pore-size distribution. For example, the critical frequency f_c in Figure 1, where there is a downward inflection in the imaginary conductivity and the real conductivity begins to flatten out, will presumably correspond to the largest pore size, or diffusion length, in the sample. Furthermore, if equation 3 is valid, then the relaxation time, corresponding to the critical frequency, will increase with the square of the maximum pore size for the sample. Moreover, because permeability is proportional to the square of pore size (e.g., Fetter, 2001), this critical relaxation time should also be a good predictor of sample permeability. For a range of sandstone samples from two nearby sites, Binley et al. (2005) demonstrate that the relaxation of their measured electrical spectra is correlated with the hydraulic conductivity. Kemna et al. (2005) also provide compelling experimental evidence of such a correlation for unconsolidated sands.

To invert complex-conductivity spectra for information about the sample microgeometry requires a specific forward model. Theoretical models have been developed in terms of grain-size distributions, pore-size distributions, and membrane polarization mechanisms (Marshall and Madden, 1959; de Lima and Sharma, 1992; Lesmes and Morgan, 2001; Titov et al., 2002; Leroy et al., 2008). However, these models are abstractions of the actual topologies and mechanisms by which ions migrate, accumulate, and diffuse within a porous medium. All of these models assume that the individual pores or grains act independently of each other, and an averaging scheme must be used to sum up the various polarization processes (e.g., Lesmes and Morgan, 2001). Furthermore, these theoretical models have numerous model parameters that can be difficult or impractical to constrain in many applications.

As an initial hypothesis, we assume that the low-frequency polarization mechanisms are diffusion controlled and that equation 3 can be used to relate a critical relaxation time (or critical frequency) to a characteristic diffusion length for the porous material. Instead of using theoretically based models to interpret the observed low-frequency complex-conductivity responses, however, we use phenomenological models, such as the Cole-Cole model (Cole and Cole, 1941), common in practical applications. These empirically derived models can fit the measured responses with fewer parameters than the theoretically derived models, although the parameters theoretically are not well defined in terms of the physical and chemical properties of the porous material.

ELECTRICAL SPECTRA MODELS

Various empirical models based on the original Cole-Cole dielectric dispersion model (Cole and Cole, 1941) have been used to fit the complex electrical response of rocks and soils. A review of existing approaches is given by Dias (2000). One of the most popular among such relaxation models is that of Pelton et al. (1978). In their model, the dispersion of resistivity is assumed to follow the same symmetric dispersion proposed by Cole and Cole (1941) for dielectric behavior.

In Pelton's Cole-Cole model, the resistivity of mineralized porous rock is given by

$$\rho^*(\omega) = \rho_0\left[1 - m\left(1 - \frac{1}{1 + (i\omega\tau_{CC})^c}\right)\right], \tag{4}$$

where ρ_0 is the DC resistivity, τ_{CC} is the time constant of the Cole-Cole model, c describes the spread of relaxation with high-frequency values representing a more peaked response, and m is chargeability (given by $m = 1 - \rho_\infty/\rho_0$, with ρ_∞ representing the asymptotic high-frequency limit to resistivity).

We express the relationship in equation 4 in terms of resistivity because it represents a model of resistivity dispersion. We could write the same expression in terms of conductivity:

$$\sigma^*(\omega) = \sigma_0\left[1 + m\left(\frac{(i\omega\tau)^c}{1 + (i\omega\tau_{CC})^c(1 - m)}\right)\right], \tag{5}$$

where σ_0 is the DC conductivity.

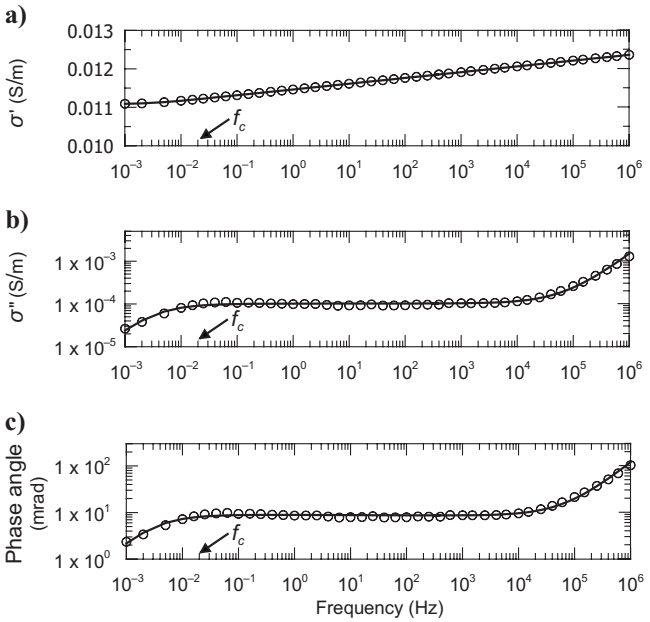


Figure 1. Example broadband electrical spectra for a sample of Berea sandstone saturated with 0.01 M NaCl. (a) Real conductivity, (b) imaginary conductivity, (c) phase angle. Symbols show measured data; solid line shows fit of generalized Cole-Cole model; f_c indicates critical frequency. Data after Lesmes and Frye (2001).

Measured electrical spectra do not always conform to the Cole-Cole model of resistivity in equation 4 as a result of two observed characteristics. First, spectra typically show dispersive behavior at low and high frequencies (see Figure 1). The low-frequency dispersion is a result of diffusive polarization mechanisms; the high-frequency dispersion often is attributed to Maxwell-Wagner polarization and is a result of physical polarization mechanisms (see Chen and Or, 2006).

Kemna et al. (2000) represent this dual polarization by using a coupled Cole-Cole model of the resistivity spectra:

$$\rho^*(\omega) = \rho_0\left[1 - \sum_{k=1}^{2} m_k\left(1 - \frac{1}{1 + (i\omega\tau_{\mathrm{CC}_k})^{c_k}}\right)\right], \quad (6)$$

where k differentiates the characteristics of the low- and high-frequency dispersion. However, the high-frequency dispersions often can be fitted adequately by adding a capacitive element in parallel with a single Cole-Cole resistivity response (i.e., equation 5), although we recognize that an equivalent high-frequency capacitive element also can be achieved with equation 6 (Kemna, 2000).

A second characteristic frequently observed in measured spectra is asymmetry in the dispersion. Davidson and Cole (1951) note this for dielectric dispersion of glycerine and propose a modification to the original Cole-Cole model. The same functional form of this model may be used for resistivity spectra (e.g., Pelton et al., 1983). Alternatively, a more generalized model can be expressed as (Pelton et al., 1983)

$$\rho^*(\omega) = \rho_0\left[1 - m\left(1 - \frac{1}{(1 + (i\omega\tau_{\mathrm{GCC}})^c)^a}\right)\right], \quad (7)$$

which is equivalent to Pelton's Cole-Cole model (equation 4) if exponent a equals unity and is equivalent to the resistivity dispersion form of the Davidson and Cole (1951) model proposed by Pelton et al. (1983) if exponent c equals unity. Pelton et al. (1983) refer to equation 7 as the generalized Cole-Cole (GCC) relaxation model. We prefer to use the term *generalized Cole-Cole resistivity relaxation* to avoid any misinterpreted equivalence to the dielectric models of Cole and Cole (1941) and others. Equation 7 is also used by Klein and Sill (1982) in their study of electrical relaxation in repacked sand-clay samples of varying grain size.

Because our study gathers data sets from samples with widely varying properties, we have adopted the generalized Cole-Cole resistivity relaxation model in equation 7 to represent the low-frequency dispersive behavior. Our investigation concentrates on spectra within the low-frequency (10^{-3}–10^{3}-Hz) range, so we prefer not to incorporate a dual relaxation model to account for high-frequency dispersion because it will not be resolved fully in this range; instead, we add a high-frequency capacitive element ε_∞ in parallel with the generalized Cole-Cole resistivity relaxation model in equation 7 to obtain (in terms of conductivity)

$$\sigma^*(\omega) = \frac{\sigma_0}{1 - m\left(1 - \left(\frac{1}{(1 + (i\omega\tau_{\mathrm{GCC}})^c)^a}\right)\right)} + i\omega\varepsilon_\infty. \quad (8)$$

The complex resistivity $\rho^*(\omega)$ is obtained by taking the reciprocal of $\sigma^*(\omega)$ in equation 8. The parameter ε_∞ represents the asymptotic high-frequency limit of the permittivity response. To uniquely define an intrinsic ε_∞ for water-bearing materials, the measurements would need to extend to frequencies on the order of 100 MHz to 1 GHz, where the permittivity response becomes independent of frequency. For the inversions in this paper, ε_∞ is essentially determined by the permittivity at the highest measured frequency, and it represents the cumulative charge storage for all polarization mechanisms which have relaxation frequencies that exceed the maximum frequency of measurement.

Figure 1 shows that the generalized Cole-Cole resistivity model in equation 8 provides a good fit to the complex conductivity of Berea sandstone measured over the frequency range of 1 mHz to 1 MHz, using only a capacitor to fit the high-frequency dispersion. The model in equation 8 provides a good fit to all of the data analyzed in this paper, and it seems that multiple relaxation models should be used only when adding a high-frequency capacitive element does not adequately fit the measured response. This makes the inversions more unique and the interpretation of the resulting model parameters more straightforward.

It is important to restate that models such as that represented by equation 8 are phenomenological; however, we can expect correlation between parameters of such relaxation models and physical characteristics of the porous medium. The relationship between the parameters of this relaxation model and physical properties of the sample is the focus here. By considering a large data set representing a wide range of physical properties, we are able to examine the wider applicability of previously proposed relationships.

We note, however, that the observed correlations between the model parameters and the properties of the porous media (e.g., time constant versus pore size) only pertain to the specific form of the Cole-Cole model used in this study. If we try a different form of the Cole-Cole model, the value of the model parameters may vary considerably (e.g., Nordsiek and Weller, 2008), and the correlations with the properties of the porous media may also differ. A significant contribution of this study is that we have collected a diverse data set, consisting of new and previously published results, and have consistently analyzed these data sets by fitting the same relaxation model to the measurements so that the model parameters can be compared directly to the physical and chemical properties of the samples. Although beyond the scope of the current work, it would be instructive to fit different forms of the Cole-Cole phenomenological model as well as theoretically based models to a diverse suite of measurements. In this way, it might be possible to evaluate the general utility and predictive skill of these different modeling approaches.

EXPERIMENTAL DATA

Sources and methods

Our aim is to compile and analyze a comprehensive data set of electrical spectra of a range of consolidated porous media. We use data from several sources (some of which are unpublished), summarized in Table 1. Common to all those investigations is that the electrical spectra of fully saturated laboratory samples has been investigated, typically in the frequency range from 1 mHz to 1 kHz, using four-point measuring cells with nonpolarizing potential electrodes. For more detailed technical information on the measurement devices, the reader is referred to the relevant publications. For all main sources, the electrical spectra were available, allowing a consistent analysis of data. In addition, key physical properties of the samples were available.

The origin of the samples and hence their composition and texture are quite diverse. Sturrock (1999), Scott (2003), Binley et al. (2005), and Breede (2006) investigate sandstone cores from the U.S.A., Germany, and the U.K. Kruschwitz (2008) also studies the spectra of a range of German sandstones, including several common (artificial) building materials such as sand-limestones, brick, and aerated concrete. We include these artificial samples to provide a wider range of physical properties in our comparison.

We include previously unpublished results, recently obtained on five more sandstone samples: a Pennsylvania Blue, a Berea, an Obernkirchner, a Gravenhorster, and an Elb. We measured electrical spectra and physical properties of these new samples. The Pennsylvania Blue Sandstone was saturated with 0.01 M NaCl (σ_w = 100 mS/m); the others were saturated with water with an electrical conductivity σ_w of 80 mS/m and an ion content similar to the synthetic groundwater used by Binley et al. (2005). Combining all of these sources results in a comprehensive data set of spectra for 50 sandstone samples and samples of six other consolidated materials. In addition, our study includes some measurements of electrical spectra on Berea and Coconino Sandstone samples saturated at different salinities to illustrate this effect in relation to the influence of physical properties.

For part of our analysis, we include data on several German sandstones (F. Börner, personal communication, 2009), some of which are reported by Börner and Schön (1991); we also include data on unconsolidated sand-clay (kaolinite) mixtures from Slater et al. (2006). In these two sources, the spectra for each sample were not available (or measured), but the studies provide supporting data on the magnitude of polarization for samples with varying physical properties. These additional sources supply data for 29 sandstones and 12 unconsolidated samples.

Table 1 summarizes our sources and key characteristics, and it shows the other available properties. The electrical spectra of all samples, where reported, were measured on samples in an orientation parallel to bedding. Mercury injection capillary pressure (MICP) measurements were used in all cases reported for determining pore-size distribution. Nitrogen BET (Brunauer et al., 1938) was used to assess S_{por}. In Table 1, σ'' is the imaginary electrical conductivity at 1 Hz (or close to this frequency) and F the formation factor.

A range of saturating fluids was used within the compiled data set. Scott (2003) and Binley et al. (2005) saturate their sandstone samples with synthetic groundwaters. Börner and Schön (1991), Sturrock (1999), and Breede (2006) use NaCl saturating fluids. Kruschwitz (2008), in contrast, uses sodium sulfate (Na_2SO_4) solutions, and Slater et al. (2006) saturates samples with NaCl brines. Details of the chemistries of the saturating fluids of the samples are summarized in Table 2. We recognize that fluid chemistry can influence the electrical spectra (e.g., Lesmes and Frye, 2001); however, we consider this effect to be minimal for the samples considered because the electrical conductivity of the saturating fluids used are similar (generally around 100 mS/m).

DATA ANALYSIS

The electrical spectra for each sample were modeled with equation 8. We used the Solver tool in Microsoft® Excel to optimize the fit, using the (unweighted) least-squares difference between model and data, based on real and imaginary conductivity. The real and imaginary components differ significantly in magnitude, so the least-squares error for each datum is scaled by the conductivity at that frequency; that is, the objective function minimized by the fitting is

$$\psi = \sum_{i-i}^{N}\left(\frac{\sigma'_i - \sigma'_{i,m}}{\sigma'_{i,m}}\right)^2 + \sum_{i-i}^{N}\left(\frac{\sigma''_i - \sigma''_{i,m}}{\sigma''_{i,m}}\right)^2, \qquad (9)$$

where N is the number of measured frequencies; σ'_i and $\sigma'_{i,m}$ are, respectively, the measured and modeled real conductivity at frequency i; and σ''_i and $\sigma''_{i,m}$ are, respectively, the measured and modeled imaginary conductivity at frequency i.

The fit results in six parameters: the DC conductivity σ_0, the generalized Cole-Cole time constant τ_{GCC}, the chargeability m; exponents a and c, and the high-frequency capacitive element ε_∞. Figure 2 shows model fits for four contrasting samples (see also Table 3 for model parameters).

Our principal focus of the modeling is to determine τ_{GCC} because this allows us to evaluate the potential generality of previously observed relationships between the time constant of relaxation models and a measure of pore-throat size. Following the earlier work of

Table 1. Sources of data used, characteristics, and key investigated properties.

	Sturrock (1999)	Scott (2003)	Binley et al. (2005)	Breede (2006)	Kruschwitz (2008)	Börner & Schön (1991)	Slater et al. (2006)	New samples
Material	Sandstone	Sandstone	Sandstone	Sandstone	Sandstone, other building materials	Sandstone	Sand-clay mixtures	Sandstone
Sample origin	U.S.A	Germany, U.K.	U.K.	Germany, U.K.	Germany	Germany, France	U.S.A.	Germany, U.S.A.
Number of samples	3	11	24	5	8	29	12	5
Frequency range (Hz)	10^{-3}–10^{6}	10^{-3}–10^{3}	10^{-3}–10^{3}	10^{-3}–10^{3}	10^{-3}–10^{3}	—	—	12×10^{3}–10^{-3}
Pore-size distribution	✓	✓	✓	✓	✓	—	—	—
S_{por} (μm^{-1})	✓	✓	✓	✓	✓	✓	✓	✓
σ'' (mS/m)	✓	✓	✓	✓	✓	✓	✓	✓
ϕ (−)	✓	✓	✓	✓	✓	✓	—	✓
F (−)	✓	(✓)	✓	✓	✓	—	—	✓

Börner and Schön (1991) and subsequent study of Slater and Lesmes (2002b) on unconsolidated samples, we also examine the influence of the pore-surface area within the sediment on the total measured electrical polarization. The polarization could be assessed using the product of the chargeability m and the DC conductivity σ_0 (e.g., Slater and Lesmes, 2002a). However, such a measure may be subject to significant uncertainty because the two parameters result from inverting measured spectra. Consequently, we consider the imaginary electrical conductivity σ'' at a particular frequency. To allow incorporation of results from Börner and Schön (1991) and Slater et al. (2006), we use the value at 1-Hz frequency.

To assign an appropriate pore-throat size for each sample (for which MICP data were available), we computed a modal pore throat D_0. The cumulative intruded mercury volume versus pressure curves were converted to a differential intrusion (in ml/g) versus pore diameter (in μm), using the Washburn equation (see Bloomfield et al., 2001). Figure 3 shows pore-size distributions for four contrasting samples (see corresponding electrical spectra in Figure 2 and tabulated values in Table 3). Note that the differential intrusion in Figure 3 is shown on a logarithmic scale to emphasize the tails of the pore-size distribution, particularly for the Pennsylvania Blue and Tennessee Sandstones.

We carried out the analysis for MICP data available for all of the samples from the studies of Sturrock (1999), Binley et al. (2005), and Kruschwitz (2008) in addition to MICP data for the new sandstone samples. For the samples of Scott (2003) and Breede (2006), we used the equivalent values of D_0 they reported. For two of the samples reported in Breede (2006) (Bu3 and Gr), MICP data were not available; however, we were able to access these samples and thus perform the MICP measurements.

TEXTURAL CONTROLS ON COMPLEX RESISTIVITY

Surface area and polarization

A relationship between σ'' and S_{por} can be considered intuitive: as the polarizable surface area increases, one would expect an increase in overall polarization. Börner (1991) demonstrates empirically such a relationship for a range of sandstones and unconsolidated sands and gravels. Furthermore, Börner et al. (1996), who assume a constant phase behavior of their samples, illustrate a significant aspect of this finding: The authors use the assumed link between surface area and hydraulic conductivity in the Kozeny-Carman model and thus reveal a potential means of estimating hydraulic conductivity through electrical-polarization measurements. Working with repacked sand-clay mixtures with a four-order-of-magnitude variation in hydraulic conductivity, Slater and Lesmes (2002b) and Slater (2007) provide further evidence of this relationship.

It is therefore useful to assess the generality of these earlier findings. Figure 4 shows the variation in σ'' (at 1 Hz) with S_{por} for the samples considered here. Included in the plot are the published results from Börner and Schön (1991) and Slater et al. (2006). A general trend of increasing polarization with surface area is evident ($r^2 = 0.68$). Several outliers from this trend are apparent — most notably, three sandstone samples with relatively high surface area.

Börner and Schön (1991) and Slater et al. (2006) do not consider the frequency dependence of σ'' to be significant and use the measurement of σ'' at a single frequency as representative for their samples. Clearly, if significant dispersion exists in the spectra, then a more robust measure of polarization would be necessary. Nevertheless, the general link between surface area and imaginary conductivity is confirmed across a wide range of natural and artificial materials. Others have conducted a similar analysis and argue that the relationship has a significant level of universality (Weller et al., 2010).

Pore-throat size and time constant

If electrical polarization in porous media occurs as a diffusive relaxation process, then we would expect a link between some characteristic length scale (e.g., grain or pore-throat size) and the time constant of a relaxation model. However, there is no evidence, to date, of any universality in such a relationship. In fact, the few studies that have explored time-constant variation in porous media reveal varying relationships. Klein and Sill (1982) prepare "artificial sandstones" from glass-bead/clay mixtures and then fit the generalized Cole-Cole resistivity relaxation model to measured spectra. Considering a range of grain sizes for purely glass-bead samples, they observe a near-linear increase of the time constant with grain size. However, measurements of spectra on glass-bead/clay mixtures reveal an increase of the time constant with increasing clay content. That is, as the overall mean grain size decreases, the length scale of the diffusion process appears to increase. The observed increase of the time constant from a 10% increase in calcium-montmorillonite clay content is similar to that seen by increasing the matrix (glass bead) grain size by an order of magnitude.

More recent studies reveal a range of experimental findings. Titov

Table 2. Fluid chemistries and conductivity σ_w of the fluids used to saturate samples. Ion concentrations in (mmol/liter).

Ion	Sturrock (1999)	Scott (2003)	Binley et al. (2005)	Breede (2006)	Kruschwitz (2008)	Börner & Schön (1991)	Slater et al. (2006)	Penns' Blue new sample	Other new samples
Cl^-	10	3.4	0.9	8.6	—	~10	—	—	1.6
NO_3^-	—	—	11.4	—	—	—	10	10	0.1
SO_4^-	—	0.3	0.9	—	8.3	—	—	—	1.2
Na^+	10	6.0	0.9	8.6	16.6	~10	10	10	1.7
Mg^{2+}	—	0.7	0.9	—	—	—	—	—	03
Ca^{2+}	—	0.6	5.7	—	—	—	—	—	2.5
K^+	—	—	—	—	—	—	—	—	0.1
σ_w (mS/m)	100	78	100	90	160	100	150	100	80.0

et al. (2002) also show evidence of relaxation length scales related to the square of grain size in sieved sands (supporting the model of Schurr [1964] in equation 3). Scott and Barker (2003) report a log-linear relationship between the frequency at peak phase angle and the dominant pore-throat size of a range of sandstones. Kemna et al. (2005) observe similar relationships in spectra from unconsolidated sand mixtures. In contrast, Binley et al. (2005) observe a linear relationship between the Cole-Cole time constant and pore-throat size. Unlike equation 3, the length scale exponent is close to unity.

Figure 5 shows the variation in τ_{GCC} of the generalized Cole-Cole resistivity relaxation model (equation 8) with modal pore-throat size D_0 for the sandstone and building-material samples compiled here. A simple relationship does not appear to exist from a quick inspection of the graph. However, most of the points on the graph do indeed follow a positive relationship between τ_{GCC} and D_0. Furthermore, the samples not conforming to this trend, with the exception of one building-material sample, appear to be those with a small modal pore-throat size (<7 μm).

Before further studying this relationship and the likely cause of any outliers, it is useful to examine other potential impacts on the time constant determined for each spectra. One limitation of the relaxation modeling applied here is the inherent nonuniqueness of the inverse solution. Such equifinality has prompted the development of stochastic approaches to studying models of relaxation processes (e.g., Ghorbani et al., 2007; Chen et al., 2008). However, the time constant is relatively well resolved in our model; we can demonstrate this examining the relationship between the position of the downward inflection of the low-frequency imaginary conductivity spectra f_c and the modeled time constant. If we express the position of this inflection in seconds as a critical relaxation time

$$\tau_c = \frac{1}{2\pi f_c}, \tag{10}$$

we can compare τ_c with τ_{GCC}, as shown in Figure 6. The strength of this relationship suggests that nonuniqueness in the parameterization of τ_{GCC} is minor for the spectra considered here. Note, however, that we do not suggest τ_{GCC} is only a function of τ_c.

We also may consider variable fluid chemistry among samples in our compilation as an influence on the electrical relaxation as a result of changes in the thickness and mobility of the diffusion paths (e.g., Klein and Sill, 1982; Lesmes and Frye, 2001; Kemna et al., 2005; Scott and Barker, 2005; Kruschwitz, 2008). To test this, we examined several sandstone samples and compared spectra at three NaCl concentrations (0.01, 0.005, and 0.002 M). The range of fluid electrical conductivity is approximately 20–100 mS/m and thus spans a wider range than that used in the compiled set of spectral data. Figure 7 shows the imaginary conductivity spectra for Berea and Coconino Sandstone samples, measured at different brine concentrations. Although both views show variation in the magnitude of polarization, there is minimal shift in the peak of the relaxation (see Table 4). Klein and Sill (1982) and Kemna et al. (2005) also report minimal difference in time constant over such a narrow range in pore-fluid conductivity. We therefore believe fluid salinity has a second-order effect on the time constant and cannot be attributed to the variation in τ_{GCC} we see in the data compiled here (Figure 5). For modal pore sizes above 7 μm, at least, variation in fluid chemistry among samples may account for some of the scatter about the general trend of increasing τ_{GCC} with modal pore-throat size D_0 (Figure 5).

For the samples with narrow pore throats, there is clearly no apparent relationship between τ_{GCC} and D_0. If we examine other physical properties of these samples, we note that these samples have a high (>20) formation factor (Figure 8) and high surface area (Figure 9) (the formation factor was not available for all published data).

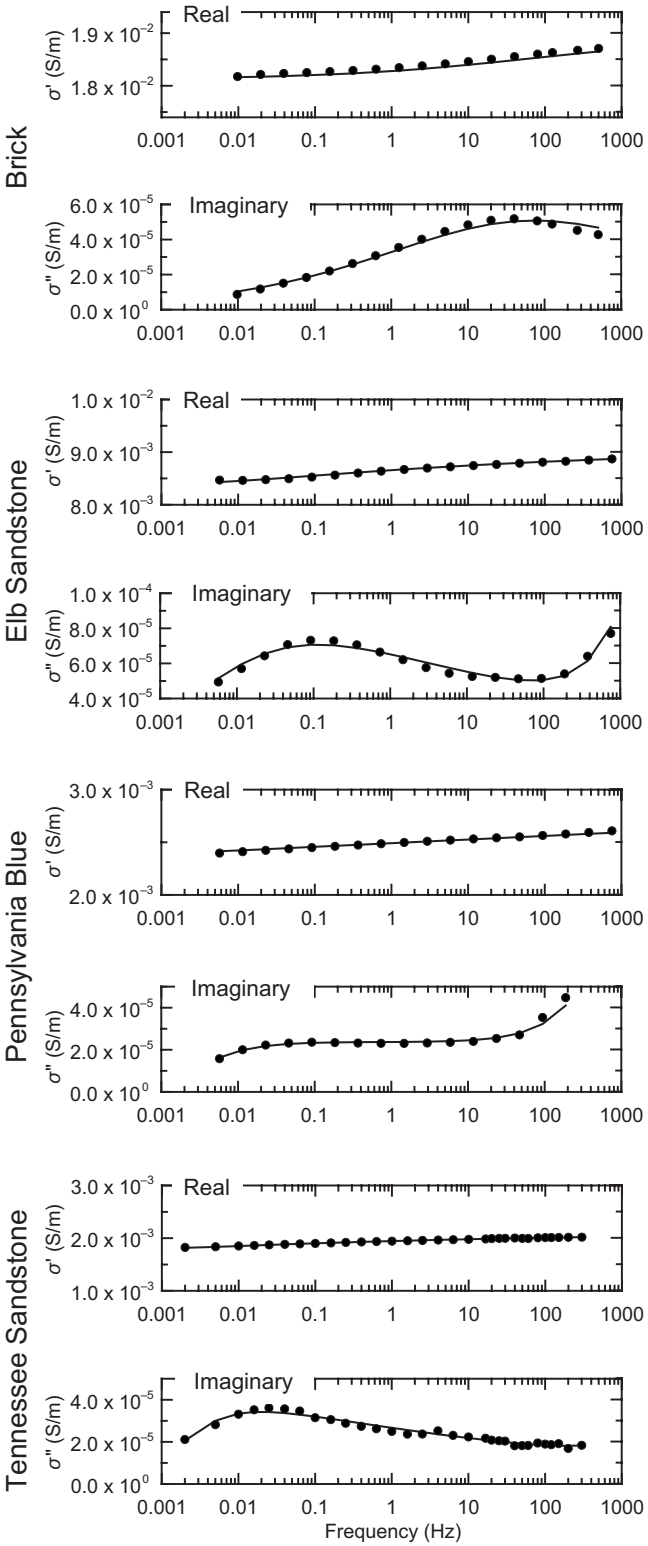


Figure 2. Measured and modeled spectra for example samples. Symbols show measured data; solid line indicates model fit. Brick data from Kruschwitz (2008); Tennessee data from Sturrock (1999).

To illustrate the trend in time constant with modal pore size, Figure 10 shows previously presented results (Figure 5) but with data symbols scaled according to the measured surface area S_{por}.

Table 3. Properties for samples in Figures 2 and 3. Note that ε serves only as a fitting parameter for the high-frequency component of the spectra; values are included for completeness.

	Brick	Elb Sandstone	Pennsylvania Blue	Tennessee Sandstone
D_0 (μm)	8.36	49.18	0.081	0.27
S_{por} (μm^{-1})	3.69	3.93	99.51	59.23
F (−)	16.6	14.0	174.0	135.0
Φ (%)	32.7	18.5	4.4	6.0
σ'' (mS/m)	0.035	0.066	0.023	0.025
σ_0 (mS/m)	18.06	8.36	2.40	1.81
m (−)	0.02	0.08	0.44	0.15
τ (s)	0.01	31.98	51.4	74.55
a (−)	0.77	0.18	0.01	0.11
c (−)	0.32	0.57	1.00	0.97
ε_∞ (F/m)	0	9.32×10^{-9}	1.50×10^{-8}	1.31×10^{-9}

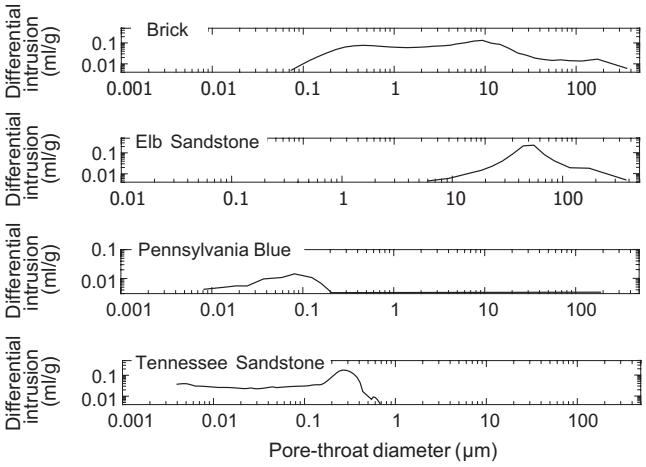


Figure 3. Pore-size distributions for selected samples.

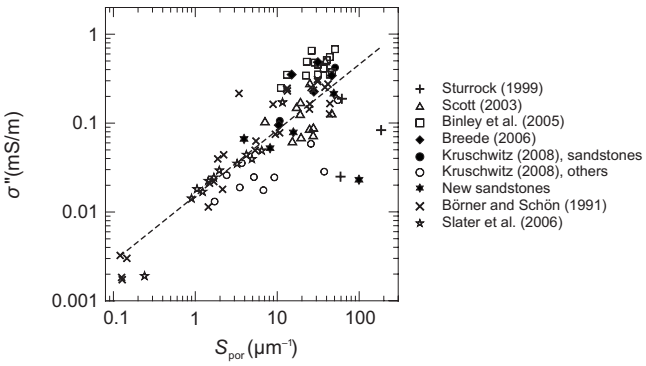


Figure 4. Variation in imaginary conductivity σ'' with S_{por} for the range of sandstone and other samples. The dashed line is the best fit of the relationship $\sigma'' \propto S_{por}^{\,n}$ ($n = 0.74$, $r^2 = 0.68$).

If we examine the pore-size distribution of these samples considered as outliers (see Pennsylvania Blue and Tennessee Sandstone examples in Figure 3), we note a distinctive (in terms of the fraction of total pore space) small pore-throat tail on the distribution, highlighting a significant proportion of very small pores — much smaller than the modal pore-throat size. It would appear that for these samples, the electrical relaxation cannot be considered to follow such a simple relationship as equation 3. The measured electrical relaxation is an integration of relaxations occurring at a wide range of scales. For some porous-media structures, a single pore-throat size may relate to a characteristic length scale for the mean time constant; for other porous media, pore throats much smaller than the physically dominant pore throat may lead to much longer electrical diffusion path lengths as they become connected, increasing the time constant. This argument may explain why Klein and Sill (1982) observe an increase in time constant with increasing clay content. For relatively coarse porous media, the polarization along such connected microstructures may be absent or may contribute little to the overall polarization. For fine-grained sediments, the polarization along these structures may dominate the observed response and thus lead to relaxation that is independent of the macropore structure (connectivity of several small pores, for example).

As surface roughness increases, one would also expect to see faster relaxation responses in the measured spectra. In fact, Leroy et al. (2008) provide compelling evidence of the effect of small-scale grain roughness on the electrical spectra of artificial porous media (made of glass grains). We believe we observe a similar effect in some of the samples we have studied; the imaginary conductivity spectra (Figure 2) for the Pennsylvania Blue and Tennessee Sandstones samples, for example, show a relatively flat high-frequency limb in the relaxation curve. Given that Maxwell-Wagner polarization cannot account for the observed imaginary conductivities in this frequency range, we attribute these observations to surface-roughness effects (shown in the pore-size distribution as very small pore throats).

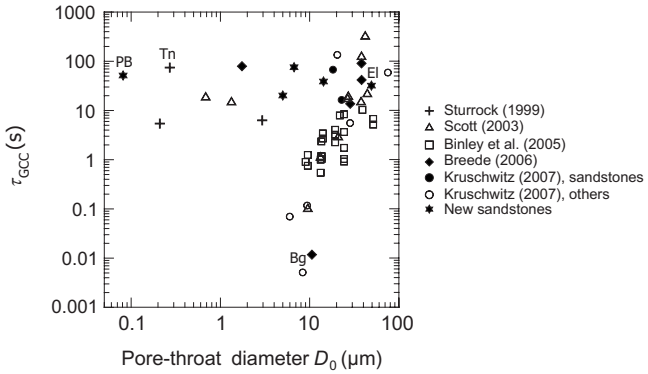


Figure 5. Variation in τ_{GCC} with D_0 for the range of sandstone and building-material samples. Specific samples referred to in the text are labeled. Legend: Tn — Tennessee Sandstone; PB — Pennsylvania Blue Sandstone; El — Elb Sandstone; Bg — brick.

If we reduce the grain size, we may observe a broadening of the electrical relaxation spectra: the small grains/pores lead to short (fast) diffusion length scales, but connectivity of many short, individual diffusion paths leads to additional long relaxation times. As these paths begin to dominate the polarization process, we see an increase in the overall mean time constant (i.e., the value determined from a Cole-Cole or similar phenomenological model).

An alternative means of interpreting the data is to consider the apparent effective diffusion coefficient D_s in equation 3 as a variable; that is, the longer-length scales of relaxation are equivalent to slow diffusion. If we adopt equation 3 as a representation of the electrical polarization process in porous media, with τ_{GCC} from our generalized Cole-Cole relaxation model as the time constant and D_0 as a measure of the effective grain diameter, then we can examine variation in D_s across the samples (Figure 11). Such analysis reveals an apparent five orders-of-magnitude variation in the effective diffusion coefficient. Differences in pore-fluid chemistry and temperature will lead to variations in the ionic mobility on the order of 10%–20% for the suite of samples. These effects can account for some of the scatter in Figures 10 and 11 but certainly not for the wide range of values and the extreme outliers. It seems that quite different polarization processes are controlling the relaxations in relatively clean and well-sorted samples with narrow pore-size distributions versus tight samples with broad pore-size distributions. For the well-sorted samples, the relaxation time increases with the modal pore-throat

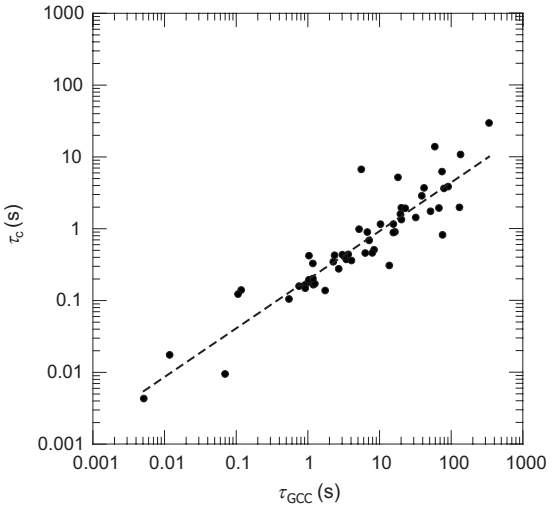


Figure 6. Variation in relaxation time τ_c (defined in equation 10) with model time constant τ_{GCC} for the range of sandstone and building-material samples. Coefficient of determination of the relationship (shown by the dashed line) between τ_c and τ_{GCC} is 0.8 ($\tau_c = 0.2\ \tau_{GCC}^{0.68}$).

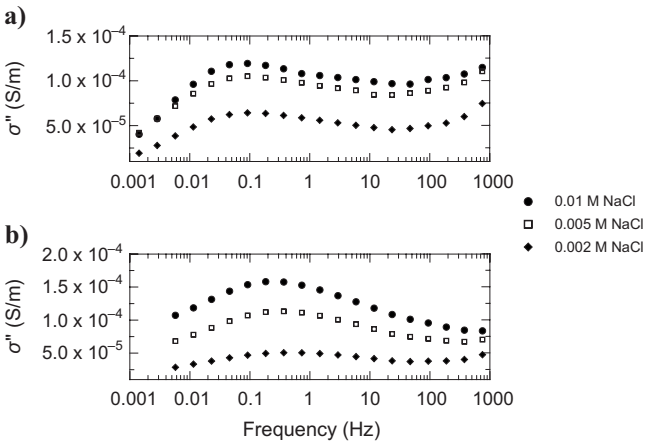


Figure 7. Imaginary conductivity spectra for (a) Berea and (b) Coconino Sandstone samples saturated with NaCl of different molarities.

Table 4. Modeled relaxation time and measured imaginary conductivity for Berea and Coconino samples and different pore-fluid NaCl concentrations.

	Berea Sandstone		Coconino Sandstone	
NaCl concentration	τ (s)	σ'' at 1 Hz (mS/m)	τ (s)	σ'' at 1Hz (mS/m)
0.002M	31.1	0.056	11.6	0.049
0.005M	51.1	0.094	10.4	0.107
0.01 M	44.6	0.106	19.8	0.146

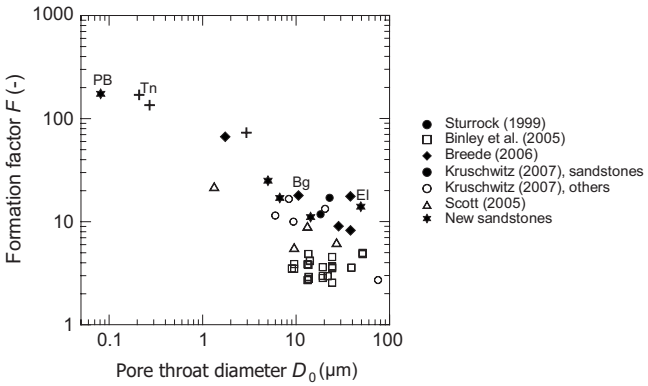


Figure 8. Variation in formation factor F with D_0 for a range of sandstone and building-material samples. Legend: Tn — Tennessee Sandstone; PB — Pennsylvania Blue Sandstone; El — Elb Sandstone; Bg — brick.

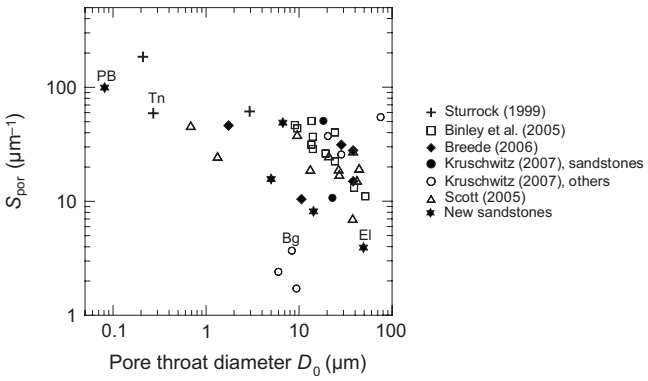


Figure 9. Variation in S_{por} with D_0 for a range of sandstone and building-material samples. Legend: Tn — Tennessee Sandstone; PB — Pennsylvania Blue Sandstone; El — Elb Sandstone; Bg — brick.

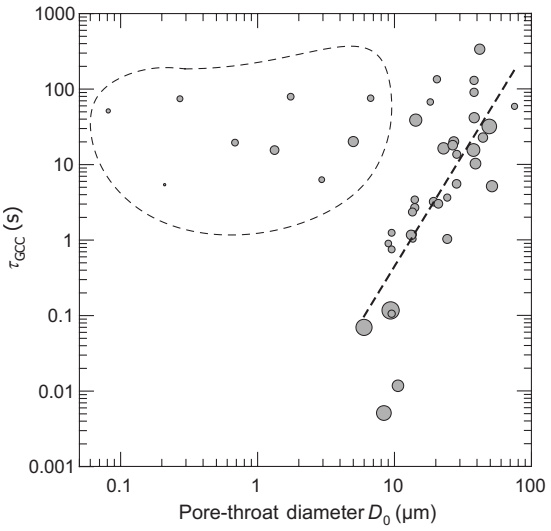


Figure 10. Variation in τ_{GCC} with D_0 for the range of sandstone and building-material samples. The bubble sizes are scaled according to S_{por} (increasing bubble size with decreasing S_{por}). The dashed line shows a best fit of a $\tau \propto D_0^n$ relationship for the data, excluding outliers marked by the broken line ($n = 2.97$, $r^2 = 0.55$).

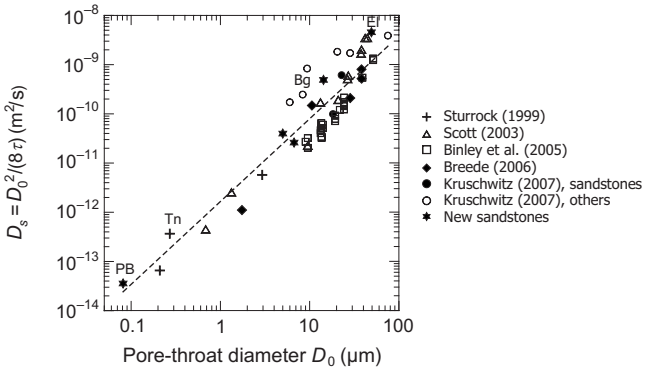


Figure 11. Variation in effective diffusion coefficient D_{eff} with pore-throat diameter D_0 for the range of sandstone and building-material samples. The straight dashed line shows a best fit of a $D_{eff} \propto D_0^n$ relationship for all the data ($n = 1.68$, $r^2 = 0.83$). Legend: Tn — Tennessee Sandstone; PB — Pennsylvania Blue Sandstone; El — Elb Sandstone; Bg — brick.

size and the apparent diffusion coefficients mainly cluster around 10^{-9}–10^{-10} m^2/s, with considerable scatter. However, for the tight rocks with small pores, the time constant is relatively insensitive to the modal pore size and the time constraint is much longer than would be expected compared to well-sorted samples with larger pore sizes.

Figure 12 summarizes the observed relationships. As discussed, the longer time constant observed for the tight samples can be explained by a slower diffusion process (i.e., smaller effective mobility from, for example, bridging EDLs between grains) or a longer effective diffusion length, perhaps resulting from the polarization of charge over multiple pore or grain lengths. Similar findings are reported by Titov et al. (2010). An important research challenge is to develop and test models, whether theoretical or more phenomenological in nature, that can robustly predict the change in polarization processes with varying pore- or grain-size distribution.

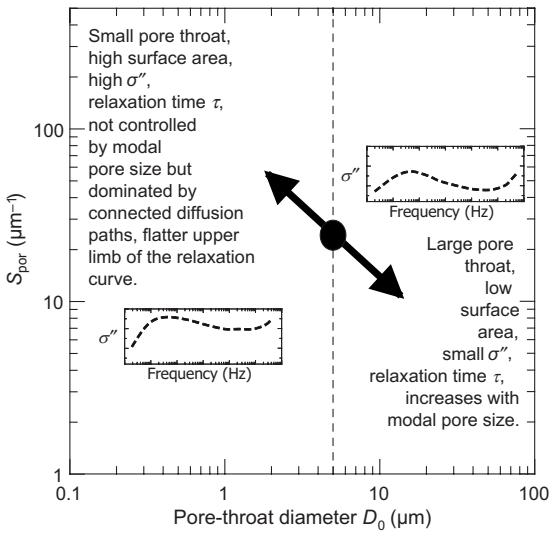


Figure 12. Summary of physical controls on electrical spectra.

CONCLUSIONS

We have examined the low-frequency electrical spectra of a range of natural and artificial porous media to assess the generality of previously proposed relationships between electrical and physical properties. Our study confirms a significant positive correlation between the magnitude of electrical polarization (quantified as imaginary conductivity at a specific frequency) and the surface-area/pore-volume ratio S_{por}. Although a degree of scatter exists within the measured relationship, there appears to be sufficient strength in the relationship to allow use of this in a predictive manner. Such use may allow estimates of S_{por} to be made in situ in the field and at larger scales than achievable through conventional laboratory analysis. Furthermore, in situ monitoring of electrical polarization may offer a means of assessing changes in surface area, resulting from artificial or natural geochemical processes.

Analyzing the parameters of a generalized Cole-Cole resistivity relaxation model fitted to many electrical spectra, we observe two apparent controls on the electrical relaxation. For samples with abundant relatively large pore throats, we observe a distinct increase in time constant with modal pore-throat size. However, for media with pore structures dominated by small pore throats, the diffusion length scales do not appear to be controlled by modal pore-throat size. We conclude that for such media, the microstructure of the network of small pores leads to some connectivity of diffusion paths and, thus, these samples exhibit relatively long relaxation times. The contributions from single- and multiple-grain/pore polarization paths may help explain previous apparent contradictory observations on artificial porous media.

As proposed by others, it appears that whereas the measure of polarization may provide a relatively robust means for estimating important hydrogeological properties, a single time constant is both influenced, to a second order, by other factors (pore-water chemistry, for example), subject to a degree of uncertainty because of the inherent nonuniqueness of model fitting and limited in the applicable range of porous media. A more general theory, possibly recognizing the distribution of relaxation times, is needed to account for the observed combined effect of the distribution of polarization length scales. This appears to be particularly important in poorly structured porous materials with relatively small pore size and broad distribu-

tions, where the relaxation time is much longer than expected from the polarization of individual pores and appears to be relatively constant and independent of the modal pore size.

Nevertheless, site- or geology-specific relationships may prove to be effective in linking electrical relaxation to properties that may be strongly influenced by pore-size distribution, e.g., hydraulic conductivity.

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